

Smart Irrigation System and Plant Health Monitoring Using Machine Learning

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Abstract—The paper presents a hybrid machine learning based method for hardware integration-based irrigation automation and potato leaf disease detection. Pre trained deep learning models extract features from leaf images then, using conventional machine learning techniques, classification guarantees correct disease identification. A soil moisture sensor continuously monitors soil conditions while a 5V DC motor controls water flow depending on moisture levels. The system combines plant health monitoring with smart irrigation control to enhance crop output. Small to medium size farmers would find this affordable option since it guarantees good water management and early disease identification.

Index Terms—IoT, Machine Learning, Soil Moisture Monitoring, Automated Plant Care, Plant Disease Detection.

I. INTRODUCTION

Water shortages and diminishing water quality are increasingly becoming issues worldwide because of increasing urbanization, industrialization and population growth and climate change [1]. From the United Nations, approximately 2.3 billion individuals reside in water stressed nations, and it is expected that this figure will grow as demand for water also grows [2]. Artificial intelligence and machine learning are new technologies that have been harnessed and applied in numerous applications and situations. One of these applications is smart irrigation, where it is essential to be capable of predicting the weather conditions to be able to ascertain the right irrigation amount for agricultural land [3]. Water use efficiency can be initially enhanced by replacing conventional surface/drip irrigation systems with contemporary irrigation techniques like sprinklers. Water serves as the main limiting element of agricultural advancement in semi arid and arid regions. It is an essential resource in boosting agricultural productivity. The advancement of agriculture is the initial step towards a civilized existence [4]. Information and communication technology has facilitated the continuous management of water

in agriculture. The use of wireless sensor networks from the crops through IoT technology and remote data management with cloud computing has led to improved monitoring of agricultural parameters that generate an enormous amount of data [5], [6].

Agriculture plays an important role in ensuring global food security, and progress in smart agricultural technologies has greatly improved crop health monitoring and resource management. One of the major challenges in modern farming is to reduce resource wastage due to diseases of the plant and the use of incompetent water. Manual water and inspection are often labor intensive, time-consuming and incredible in traditional farming operations. A self-sufficient, real-time plant care system that addresses these issues is made up of a combination of automatic irrigation control with machine learning-based disease diagnosis in machine learning-draped plant health monitoring with smart irrigation.

This system captures potato leaf images, analyzed using KN, Random Forest and SVM machine learning models, such as potatoes quickly to identify diseases such as blight, potato late blight and healthy potato leaves. Of these, SVM provided the highest classification accuracy. To determine the water requirements of the plant, a soil moisture sensor also constantly evaluates the condition of the soil. Depending on moisture content, an RGB LED visually displays the condition of the impaired soil. High water flow is activated by low water flow by red (dry soil), blue (optimal moisture) and green (adequate moisture) by medium water flow. Arduino UNO acts as the main controller, manages the sensor data and controls the water pump to ensure the development of the optimal plant.

By detection of machine learning-based disease and taking advantage of automatic irrigation, this system increases accurate agriculture, uses resources, and reduces manual labor. The proposed solution aims to improve crop productivity, increase plant flexibility.

II. RELATED WORK

This paper presents an IOT-based plant disease detection system that integrates machine learning for real-time monitoring [7]. This plant employs image processing techniques to analyze leaf health and detect diseases in the initial stage. The system uses IOT sensors to collect environmental data, which aids adaptation of plant care. By taking advantage of smart technology, the system increases agricultural productivity, reduces manual intervention, and reduces resources wastage.

The study states how ML and IOT contribute to accurate agriculture and sustainable farming. This research proposes deep learning for accurate diagnosis and accurate treatment and proposes a disease management system integrating IOT [8]. Convolutional neural networks (CNN) classify plants based on leaf images, which reveals quickly. IOT sensors constantly monitor soil moisture and environmental conditions, automate irrigation and prevention of disease. System reduces manual labor and optimizes farming processes, which improves resource use and crop yield. The study emphasizes the importance of integrating AI and IOT in accurate agriculture.

This paper explores an IOT-and AI-operated system to detect the shortcomings of crop diseases and nutrients in the fields on a large scale [9]. It integrates the image-based disease classification and deep teaching models for IOT sensor for real-time monitoring of soil and environmental conditions. The system helps in detecting initial disease, reducing crop loss and improving yield. By reducing manual labor and ensuring accuracy in agricultural operations, studies have been exposed in the study how AI and IOT technologies increase farming efficiency and promote permanent agricultural practices. This study presents an image processing and deep learning models to detect and classification of plant disease [10]. Using CNN architecture, the system plant extracts the features from the plant leaf image and classifies diseases with high accuracy. The paper highlights the role of AI in accurate agriculture, where the detection of automatic disease reduces dependence on manual inspections. Research suggests how deep education enhances agricultural monitoring, which improves plant health management and customized farming strategies for better productivity and stability.

This paper introduces a better Yolov5 deep Learning algorithm to detect the defect in wood processing [11]. Although it is not directly related to agriculture, its approach can be applied to detect the disease to facilitate extraction by using the attention system. The study demonstrates how modifying a deep learning model improves detection accuracy and efficiency. The working plant provides useful data for enhancing the AI model for health monitoring and illustrates how machine learning developments in several fields may facilitate precision farming.

The study presents MDfC-RSNET, a deep learning-based system, which is integrated with IOT for accurate crop disease recognition [12]. MDfC classification improves accuracy. IOT sensors collect environmental data, which increases the prediction and prevention strategies of the disease. Paper has

been highlighted how IOT and AI can increase agricultural productivity and reduce loss in agriculture. In addition to reducing dependence on chemical remedies and guaranteeing maximum resource usage in smart farming, this technique provides farmers access to real -time crop health information. This research presents the field plant, which is a large dataset of field plant images, designed to train deep learning models to detect plant disease [13]. Dataset is important for improving CNN-based architecture used in accurate agriculture. The study highlights the importance of high- quality dataset in AI-operated plant health monitoring. By offering a standardized dataset, research supports progress in machine learning applications in agriculture, improves the accuracy and reliability of the system detection system

This paper reviews the application of deep teaching techniques, especially firm nervous networks CNN for planting and classification of plant disease [14]. This offers intensive analysis of intensive analysis of various CNN architecture, their effectiveness in processing plant leaf images and challenges such as dataset quality, computational cost and model adaptation. The review highlights the importance of deep learning in accurate agriculture, improves disease detection accuracy, reduces manual intervention and enables real -time monitoring. It also discusses future progress in intensive learning, increases automatic agricultural practices for stability and efficiency. This paper focuses on the development of fire-fighting robots using deep learning and machine vision for object detection [15]. The system employs the CNN for real -time image analysis to identify and classify objects, which is capable of taking autonomous decisions in fire-fighting operations. While mainly designed for firefighting, the technology used, such as object recognition and convenience, can be applied to the disease detection. The study shows how intensive learning can be used for automatic monitoring, increasing accuracy in various domains including agriculture.

This study examines the use of CNN for extraction of texture features for classification of leaf diseases [16]. By focusing on separate textual patterns of both healthy and sick leaves, the study enhances the ability of deep learning models to differentiate between diseases of different plants. This letter explains how an increased CNN architecture increases accurate agricultural efficiency and increases the accuracy of the disease detection. Modern machine learning techniques can automate health monitoring, reduce the need for physical inspection, and increase agricultural productivity displayed by the research center.

III. DISCUSSION

Agricultural plays a crucial role in worldwide meals production, and development in agricultural era has substantially improved crop management and useful resource usage. One of the number one challenges in contemporary farming is to save you resource wastage throughout efficient water control and early disease detection. Traditional farming techniques rely an excessive amount of on guide inspection and irrigation,

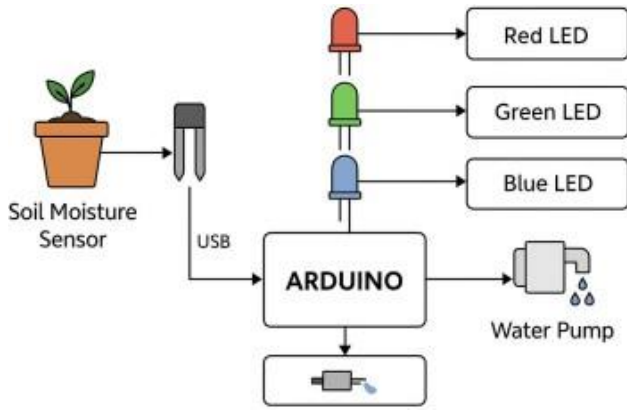


Fig. 1. Arduino Hardware Block Diagram

which may be time ingesting, inconsistent and hard work-intensive. To resolve those challenges, the undertaking introduces a machine getting to know-based plant health tracking system integrated with an automatic irrigation mechanism. The machine is designed to investigate plant fitness, detect illnesses via the usage of system learning fashions and alter irrigation based totally on actual -time soil moisture levels, which can reduce manual labor and adapt to agricultural productiveness.

The system workflow starts with the purchase of potato leaf photographs, which might be analyzed the use of K-Nearest Neighbors (KNN), Random Forest, and SVM models. Among the ones, SVM examined the fine elegance accuracy, efficaciously distinguishing between Potato Early Blight, Potato Late Blight, and Healthy Potato Leaves. By identifying plant illnesses at an early degree, the tool allows timely intervention to prevent sickness unfold and reduce crop damage. Simultaneously, a soil moisture sensor constantly video show units soil situations and transmits actual-time records to Arduino UNO, which serves because the treasured processing unit. Based on the moisture stages, the tool automatically controls the irrigation mechanism the use of a 5V DC motor-pushed water pump, ensuring gold popular water distribution. To offer actual-time comments on plant situations, an RGB LED visually suggests the plant's popularity, helping farmers quick look at plant health without requiring non-stop manual monitoring.

One of the maximum important factors of plant health monitoring is the device's trouble with picture processing and disease detection. Immoderate-desire images of potato leaves are taken as the initial segment in the procedure, and they go through some of preprocessing steps to enhance the accuracy of the evaluation. These strategies include noise bargain to remove distortions, shade enhancement to modify brightness and assessment, and segmentation to spotlight affected regions of the leaf. The processed pics are then fed into the device studying fashion, in which the SVM plant performs better than KNN and random forest in classifying fitness conditions. By analyzing the gadget to detect the disorder, the gadget ensures

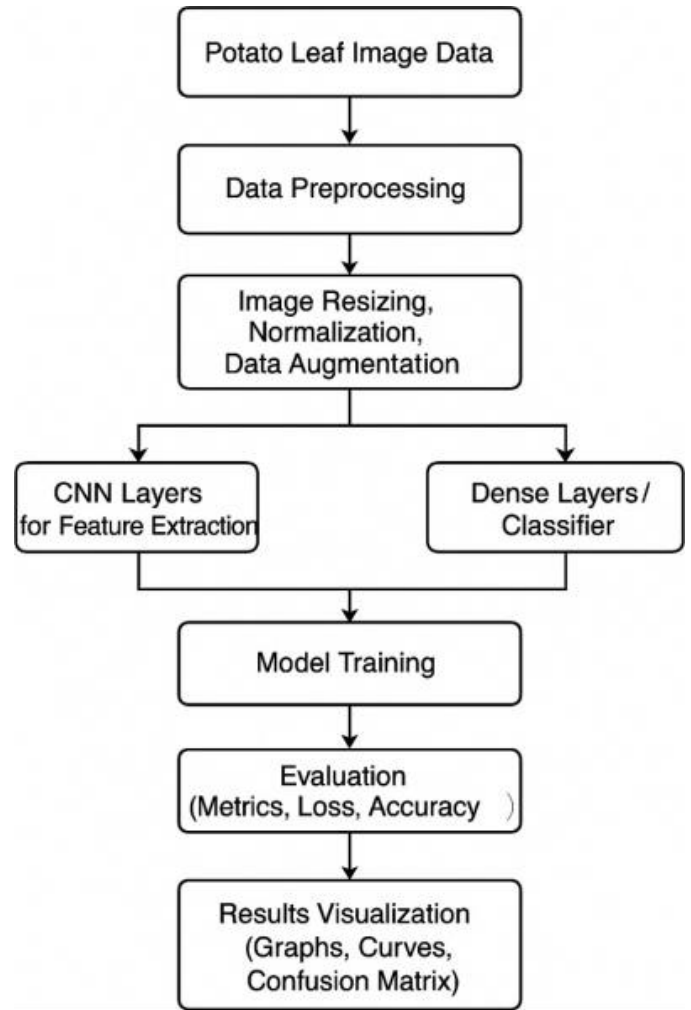


Fig. 2. Machine Learning Workflow

accurate type, reduces dependence on human inspection, and enables early disease analysis for advanced crop protection.

Together with sickness detection, soil moisture evaluation and automated irrigation tool feature to optimize water use and hold ideal ground situations for plant increase. The soil moisture sensor continuously measures soil moisture ranges and sends that information to the UNO Arduino, which determines the plant's irrigation needs. The machine operates on a 3-tier moisture splendor mechanism, in which an RGB LED acts as a visible indicator of soil situations. If the soil is in fact too dry, the LED turns red, and the system turns on immoderate water go with the float. If the soil has optimal moisture, the LED turns blue, and water flow with delay in loop is induced. If the soil moisture is enough, the LED glows green with no water flow. This actual-time seen signaling ensures inexperienced tracking of plant health and irrigation needs without requiring everyday manual supervision.

IV. RESULTS

The integration of system gaining knowledge of plant-based illness detection and automatic irrigation manipulate

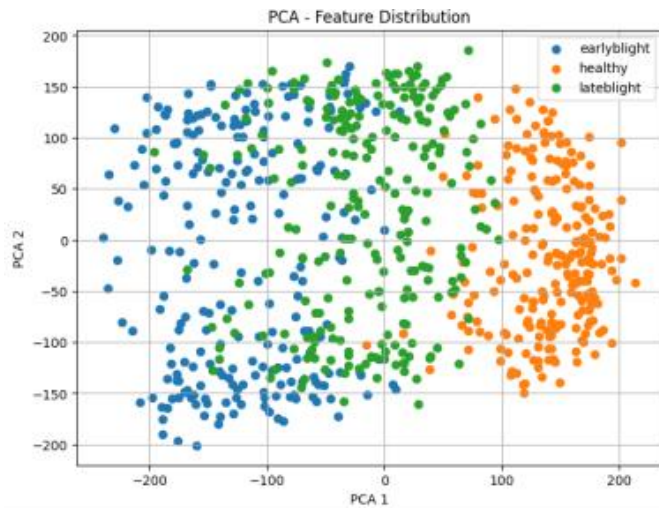


Fig. 3. PCA

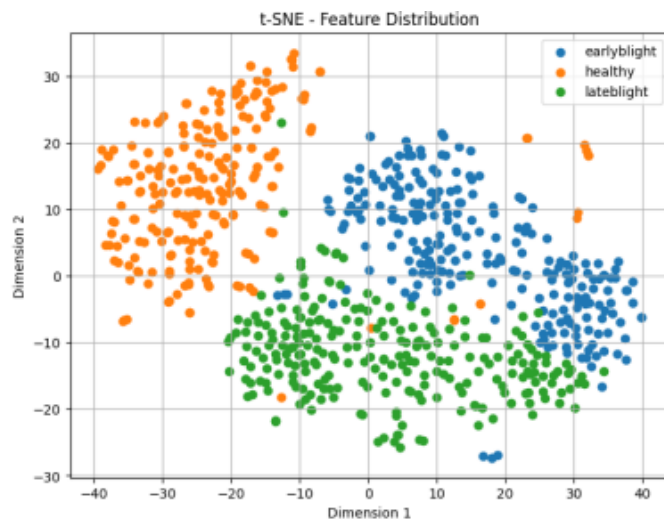


Fig. 4. t-SNE

significantly improves agricultural trendy average overall performance. By making use of SVM for specific disease identification and a soil moisture sensor-pushed irrigation mechanism, the tool ensures real-time tracking, reduces guide tough art work, and optimizes water usage.

V. KEY LEARNINGS

This challenge shows how system reading and automation can enhance farming via detecting plant illnesses and coping with irrigation effectively. By evaluating SVM, KNN, and Random Forest, SVM have become positioned to be the maximum accurate for figuring out potato leaf diseases. A soil moisture sensor helps reveal water ranges, and based totally in this information, an RGB LED modifications shade to signify soil conditions. The Arduino UNO acts as the main controller, adjusting the 5V DC motor to manipulate water flow. This



Fig. 5. Disease Classification of Plant Leaf using SVM

Class	Metric	KNN	Random Forest	SVM
0	Precision	0.86	0.9	0.98
0	Recall	0.98	0.92	0.98
0	F1-Score	0.91	0.91	0.98
1	Precision	0.96	1	1
1	Recall	0.96	0.97	1
1	F1-Score	0.96	0.98	1
2	Precision	0.95	0.87	0.98
2	Recall	0.79	0.89	0.98
2	F1-Score	0.87	0.88	0.98
-	Accuracy	0.92	0.93	0.99
-	Macro Avg Precision	0.92	0.92	0.99
-	Macro Avg Recall	0.91	0.92	0.99
-	Macro Avg F1-Score	0.91	0.92	0.99
-	Weighted Avg Precision	0.92	0.93	0.99
-	Weighted Avg Recall	0.92	0.93	0.99
-	Weighted Avg F1-Score	0.92	0.93	0.99

TABLE I
PERFORMANCE COMPARISON BETWEEN MACHINE LEARNING MODELS

device reduces manual work, saves water, and helps healthier crop increase, making farming more efficient and sustainable.

VI. COMPARISON WITH EXISTING WORK

REFERENCE	ACCURACY	PRECISION
[16]	96.14%	—
[17]	87.68%	—
[18]	98.33%	90.5%
[19]	96.5%	93%
[20]	99.99%	99.9%
[21]	97%	97%
Proposed Work	99%	99%

TABLE II
PERFORMANCE COMPARISON BETWEEN EXISTING LITERATURE

VII. CONCLUSION

This system offers an automatic plant fitness monitoring and smart irrigation device that complements agricultural efficiency. By utilizing tool studying literature, plant illnesses are detected with immoderate accuracy, with SVM proving to be the handiest in most of the evaluation algorithms. A soil moisture sensor continuously video display devices moisture levels, ensuring maximum satisfactory irrigation manage thru a 5V DC motor. The RGB LED signs offer instant comments on soil conditions, permitting timely interventions. This machine minimizes manual hard work, optimizes water usage, and improves crop health, contributing to sustainable and inexperienced farming practices.

VIII. FUTURE SCOPE

Machine learning-powered plant health monitoring with smart irrigation system has immense ability to progress for future, making it a promising solution for accurate agriculture and durable farming. An important area of improvement is increasing the machine learning model by integrating advanced deep learning techniques such as you look only once or vision transformer to detect more accurate and sharp disease. The system can also be expanded to detect a wide range of plant diseases by training models with a large dataset covering

many plant species. Additionally, integrating the cloud-based database may enable remote monitoring, where farmers can use real-time plant health and soil moisture data from their smartphone through mobile application. Future development may involve involving weather forecast models, allowing the system to adjust irrigation based on environmental conditions, allowing water use to be more customized. The introduction of automatic pesticides can only increase the safety of plants by applying treatment, reduce chemical use and promote environmentally friendly farming.

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SMART IRRIGATION AND PLANT HEALTH MONITORING USING MACHINE LEARNING

MICROPROCESSORS AND MICROCONTROLLERS

COURSE CODE – BECE204L

SLOT - D1+TD1, L49+L50

Submitted by:

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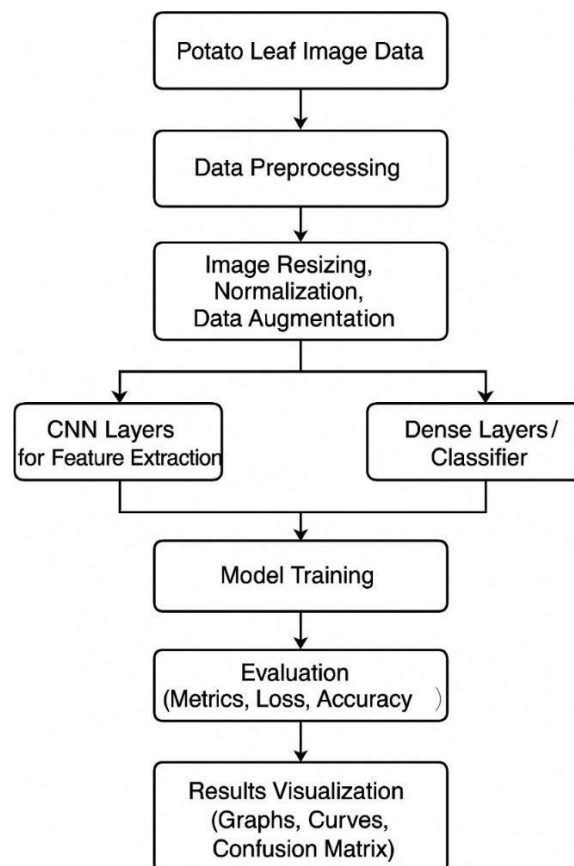
Aim

The principal intent of the selected project has been to formulate an intelligent irrigation system with some plant health monitoring. With the intelligent irrigation, there is efficient utilization of water by being able to detect soil moisture and to provide just the right amount of water at the right time. The other part of the plant health module includes machine learning techniques for detecting diseases based on image classification of potato leaves. This integrated system aims to address two fields in agriculture; resource management and crop safeguard. To save water and help farmers act promptly against diseases.

Such a system should ideally intervene minimally and enhance the consistency of plant health, thus achieving a rise in productivity with the different farming sizes in mind. Automating and utilizing artificial intelligence make this system see the future of smart agricultural technology.

Block Diagram

Below is the block diagram representing the structure of the smart irrigation and plant health monitoring system:



The framework for our intelligent system can now be revealed in the form of a block diagram. The core part of this system is the image collection module, which contains healthy and unhealthy leaf records of images. Later, these images are subjected to preprocessing before using a Convolutional Neural Network (CNN) for disease detection. Along with this, moisture content is acquired through a sensor and then processed via an Arduino microcontroller. The moisture level detected through the microcontroller activates or deactivates a water pump through a relay. The LEDs are used for the display of soil moisture: Green moist, blue moderate, and red dry. Finally, the system gives outputs on disease detection an irrigation system in terms of graphs, metrics, and real-time actions. Hardware combined with software enables real-time monitoring of farms and smart decision-making, hence a total solution for smart farming.

Description about Blocks

1. Potato Leaf Image Data - We begin by gathering a dataset of potato leaf images, both healthy and diseased.
2. Data Preprocessing - The initial segment of the system involves capturing potato leaves (which are either healthy or suffering only early blight or late blight) and is taken from the plant. Those images are integral to train the machine learning model. From there, the dataset goes into the pre-processing phase to enhance the photos quality and remove other areas of the image that don't relate to the main spectacle and could distract the model. The noise was removed using smoothing techniques, and the primary features of the leaf were highlighted by adjusting contrast and brightness levels. The background either blurred or removed so that the model would only pay attention to the leaf. This stage of pre-processing ensures the model's representation of the images is clean, crisp, and uniform in quality and consequently, has an effect on the model's accuracy. Doing proper re-processing gives the machine learning pipeline a solid start that prevents the model from classifying ambient noise and concentrate on the key identifiers.
3. Image Resizing, Normalization, Data Augmentation and CNN Layers for Feature Extraction - The first step of the system is to capture images of healthy and infected potato leaves. These images are critical for training of the machine learning model. Once the images have been captured, the dataset will then need to be pre-processed to improve the quality of the images, and remove any excess from the image that could cause confusion in the model. The extra noise was removed using smoothing techniques, while important parts of the leaf were enhanced through contrast and brightness methods. The background was blurred or removed entirely, to ensure that the model only model the leaf, and not any other extraneous information. This step ensures that the images can be displaced, they will be sharp and of consistent quality, and it will affect the model's performance as it trains. With good pre-processing you create a sound foundation for a strong machine learning pipeline, and instruct the model not to model irrelevant noise, but to model the important features. Normalization is used to change the pixel values display for improved learning performance. Furthermore Data Augmentation (rotate, flip, etc.) is used to create variability within the dataset, for more training, and helps to prevent overfitting.
4. Dense Layers / Classifier and Model Training - The subsequent phase would involve feeding the inputs to the deep layers into a neural network to perform classification. The soft max activates these layers which record the probabilities that could be associated with each illness class concerning the defined spotted pattern. This is where a loss function comes into play through training; it records how projected categories deviate from true labels. Adam or SGD is generally employed in optimization under the optimizer category, modifying the weights of the network to result in minimums of the error as much as possible. In between, in order to avoid overfitting, several training epochs will be followed once the data are split as training and validation sets. Thus, improving the classification accuracy and reducing the loss are every training's main motives. The training collects inputs in a batch while efficiencies of different epochs are recorded.

5. Evaluation (Metrics, Loss, Accuracy) and Results Visualization (Graphs, Curves, Confusion Matrix - Following the training process, assessment viably estimates the performance of the model in real-life situations, employing a dataset that is kept freed from intervention in the initial computing configurations for that aim. Various metrics for estimation are calculated like accuracy, precision, recall, and F1 score. Using precision, we can assess the correct predicted positives, while recall gives us the actual counted positives found. Performance among different types of diseases, accurate as well as misdiagnosed disease types, can be pictorially represented using confusion matrices. The graphical representations on the loss in relation to the epoch and accuracy over time show the learning pattern of the model. Not only that, it tells, through the ROC-AUC curve, how well discriminating classes is done by the model. This upward trend will thus not only ascertain the effectiveness of the said model but also ensure further optimal efforts. Visualization turns simple metrics into intuitive metrics for both researchers and farmers.

Flowchart of the Code

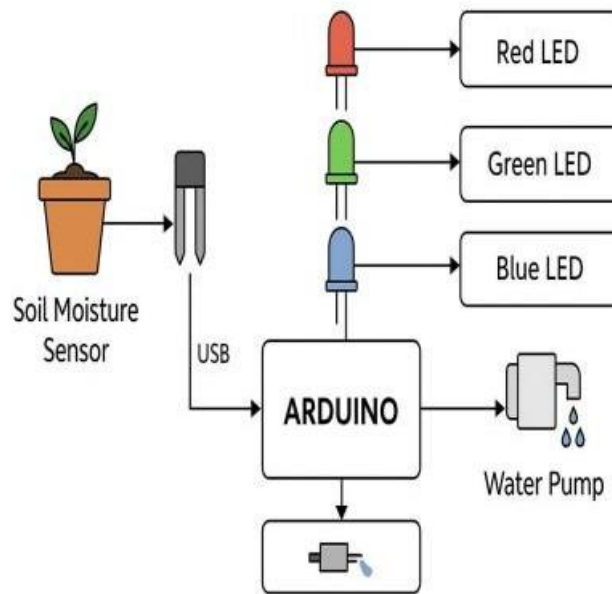


Fig. 1 Flow chart of machine learning work flow

This figure shows the entire machine learning process for identifying diseases in plants. Actual Data Collection refers to a stage that involves collecting huge datasets of annotated leaf images. These images will then go to Preprocessing, which ensures that uniform quality and size are maintained for every processed image. The processed data is, thus, fed into the CNN Model that is constructed with layers like convolution, pooling, and dense layers. Model Training is understood to be where knowledge is acquired through loss minimization from labelled data. Later, in the Assessment stage, it will be evaluated with new data to determine performance metrics. It will finally get connected to a smart irrigation system capable of acquiring real-time data through the ESP32 camera and analysing the leaf along with the data capture. Such a solution provides ongoing improvement and simple scalability, thus becoming a sustainable option for intelligent farming.

A complete water colour painting showing-the whole process of machine learning for plant disease detection. Starting with Data Collection, it involved large data sets on annotated leaf images. These images will then go through preprocessing, ensuring that all of them have uniform quality and size. The processed data is then fed to this CNN Model constructed with layers such as convolution, pooling, and dense. It is training with the help of their labelled data and corresponding competition of loss. Later it was assessed using new data in the Assessment stage, which would determine the performance metrics. The model's final outcome serves as the interface to the smart irrigation system that captures real-time data using the ESP32 camera while also analysing leaf health concurrent with data capture. Therefore, it is an ever-evolving, easily extending, sustainable solution for intelligent farming.

Literature Review

The contemporary research of smart agriculture emphasizes the usage of Support Vector Machines (SVM) or Convolution Neural Networks (CNN) for plant disease classification. Studies suggest that coupling Machine Learning techniques with IoT would significantly enable better resource management and diagnosis of plant health. However, very few have real-time irrigation automation in their systems. Therefore, the present work combines disease identification with automatically controlled irrigation. Models from various IEEE, Springer, and agricultural research journals have been evaluated for improvement to the design. The system is smart and practical because our method combines software-based disease classification with hardware-based water delivery. Compared to traditional systems, our solution is more accurate, provides instantaneous feedback, and requires less human intervention, making it convenient for contemporary precision agriculture. The dual module in this system sets a precedent for all future farm technology applications.

An intelligent irrigation system is a real-time, fully automated system that works on the foundations of Arduino. It begins with a Soil Moisture Sensor that measures the moisture content in the soil. After the moisture information has been fetched by the Arduino, it processes and does further action. Accordingly, moisture levels operate through its Relay Module to control the Water Pump for either switching it on or off. USB cable connecting the Arduino to a laptop provides power. Three-colored LEDs show the soil condition in the system, namely, green for wetness, blue for medium moisture, and red for dryness. Prompts can be provided according to what's happening in the soil environment; instant feedback of soil conditions can make it really convenient for the user since it does not take long time for them to know the soil conditions. This will allow intelligent irrigation planning which corporates efficient water use and healthy plants and can be used in small backyard homesteading or greatly expanded areas of agriculture.

It's based on soil moisture content, if soils are arid then the red LED lights on, and the water pump functions for 12 seconds, dispensing about 15mL of water; while in humid soil, the blue LED is lit while the pump runs within 8 seconds and releases around 10mL of water. Whereas, if the soil has moisture present, then it would glow the green lamp and the pump would run only for 4 seconds discharging 5mL of water into it. Simple but effective reasoning guarantees proper utilization of water supplied to plants. All components are connected on a breadboard with one wire per slot to eliminate interference; no external battery or transistor is used, just core modules for simplicity. The result is a very precise, completely automatic, economical irrigation system to be used in agricultural fields.

Discussion on Pros and Cons

The incorporation of machine learning and embedded systems design is mainly directed toward creating smart solutions for agriculture. Its focus is primarily on irrigation events as well as the detection of leaf diseases, which enables accurate conservation of water and timely treatment of diseases.

Pros:

- This would be greatly effective in real-time soil moisture monitoring.
- CNN-based precise evaluation of the diseases.
- It saves water and reduces the human effort.
- Affordable and easy to implement.

Cons:

- Limited based on a specific potato leaf dataset.
- Accuracy of the sensor is the key factor.
- Needs constant power and internet connectivity for online functionality.

Result of test image

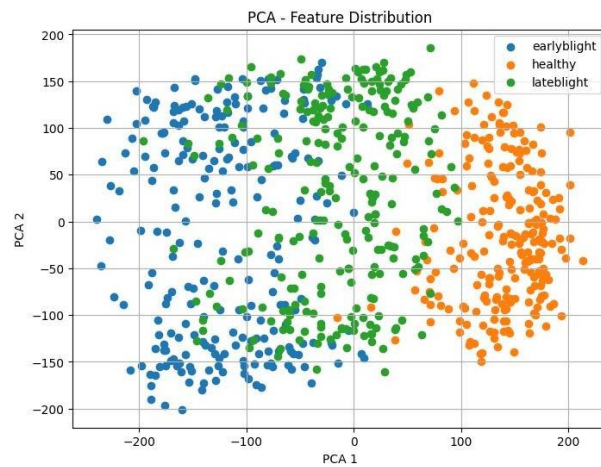


Fig. 2 PCA Feature Disribution

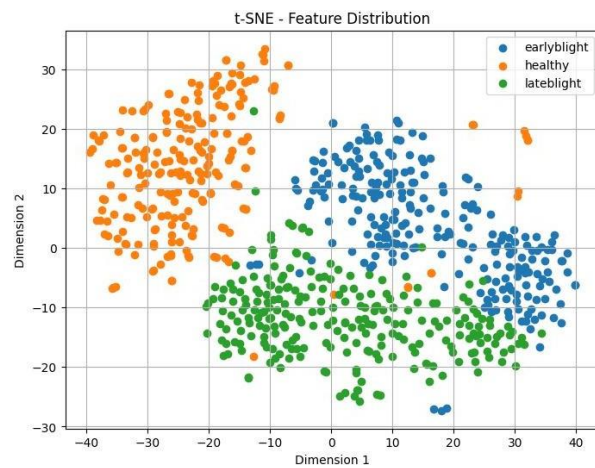


Fig. 3 t-SNE Feature Distribution

Testing: 1.JPG



Predicted class: earlyblight

Testing: 2.jpg



Predicted class: healthy

Testing: 3.JPG



Predicted class: lateblight

Class	Metric	KNN	Random Forest	SVM
0	Precision	0.86	0.9	0.98
0	Recall	0.98	0.92	0.98
0	F1-Score	0.91	0.91	0.98
1	Precision	0.96	1	1
1	Recall	0.96	0.97	1
1	F1-Score	0.96	0.98	1
2	Precision	0.95	0.87	0.98
2	Recall	0.79	0.89	0.98
2	F1-Score	0.87	0.88	0.98
-	Accuracy	0.92	0.93	0.99
-	Macro Avg Precision	0.92	0.92	0.99
-	Macro Avg Recall	0.91	0.92	0.99
-	Macro Avg F1-Score	0.91	0.92	0.99
-	Weighted Avg Precision	0.92	0.93	0.99
-	Weighted Avg Recall	0.92	0.93	0.99
-	Weighted Avg F1-Score	0.92	0.93	0.99

Table. 1 Performance comparison between machine learning models

REFERENCE	ACCURACY	PRECISION
[16]	96.14%	—
[17]	87.68%	—
[18]	98.33%	90.5%
[19]	96.5%	93%
[20]	99.99%	99.9%
[21]	97%	97%
ProposedWork	99%	99%

Table. 2 Performance comparison with existing literature

The outcomes are

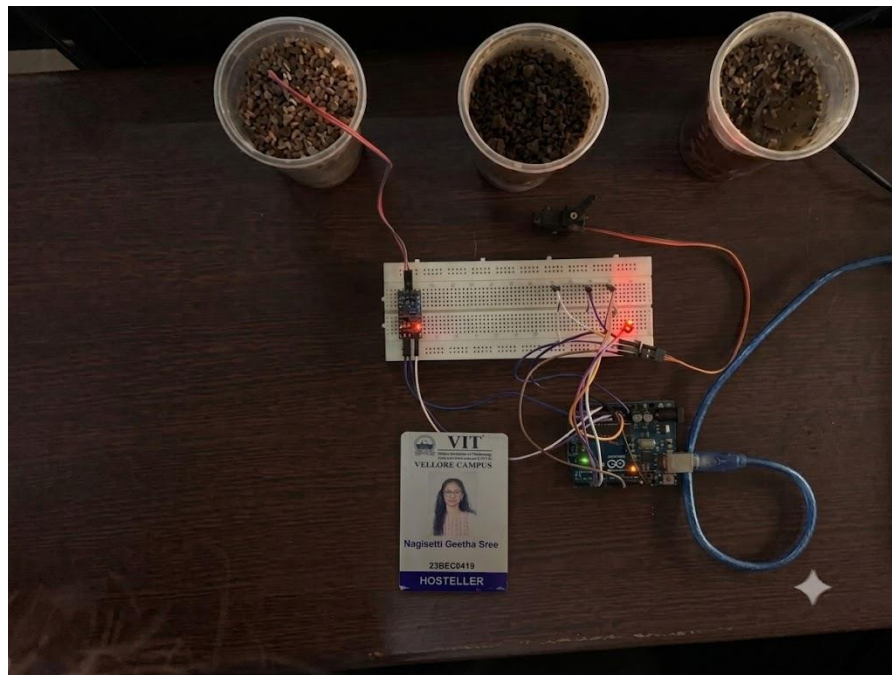


Fig. 3 Outcomes of dry conditions

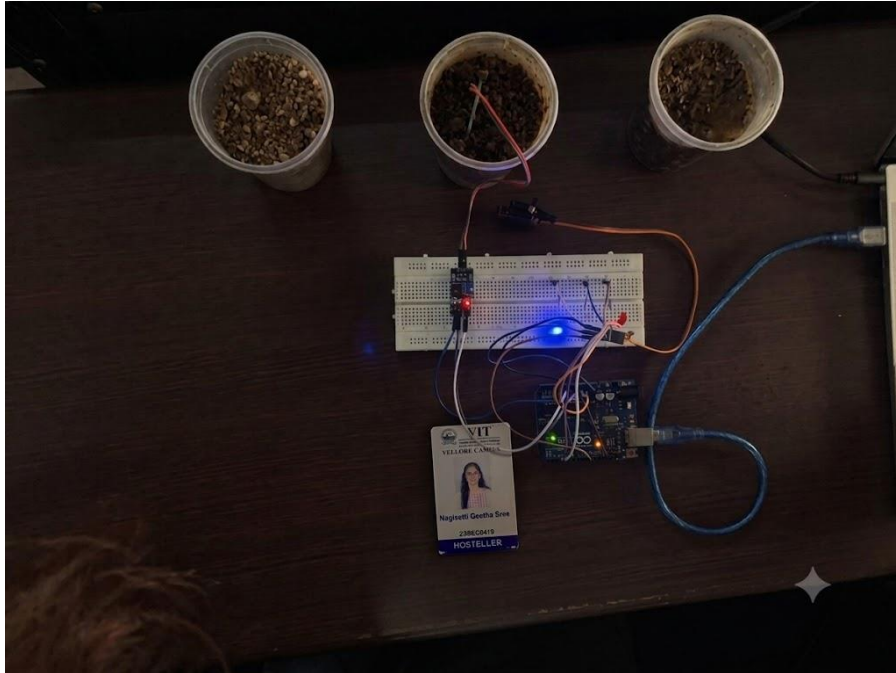


Fig. 4 Outcomes of optimal conditions

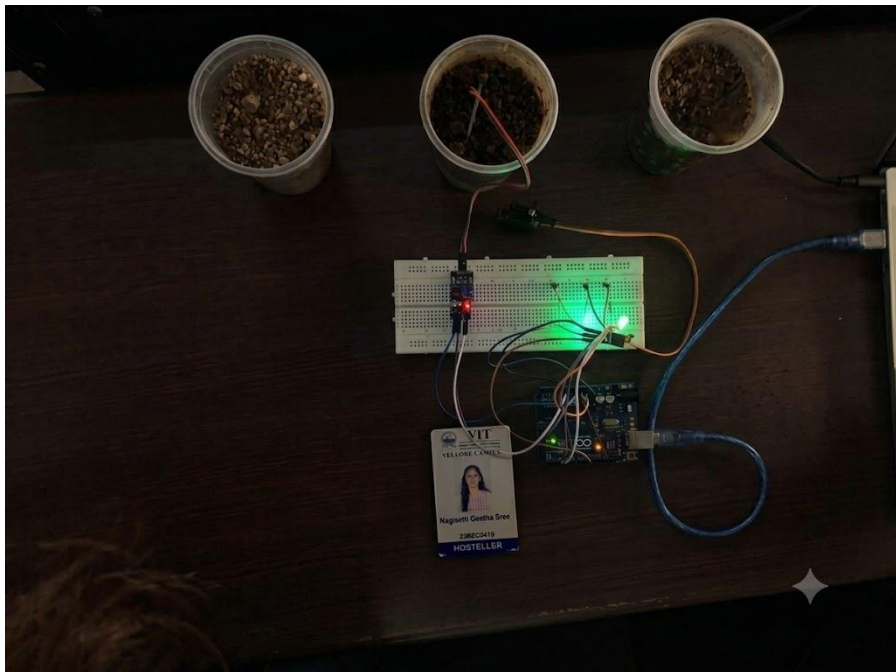


Fig. 5 Outcomes of moist conditions

